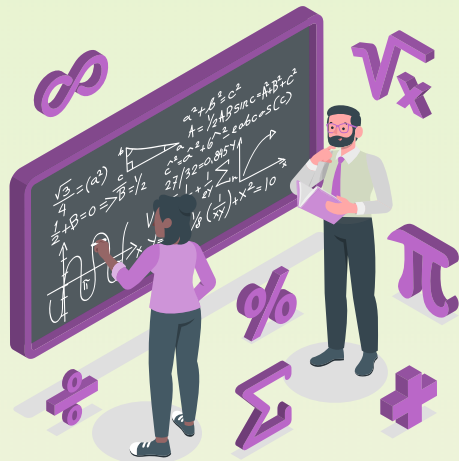




# LINEAR EQUATIONS IN TWO VARIABLES

## Linear Equation

- An equation in which the power of variable is one, is called a linear equation.

**For eg.**

$$4x + 5 = 3x + 1$$

$$2x + 3y = 45$$

## Solution of a linear equation in one variable

- Value of the variable which when substituted in equation, makes two sides of the equation equal, is called the solution of the equation.
- For e.g. Consider the equation  $2x + 7 = 0$ . Its solution i.e. the root of the equation is  $-7/2$ . This can be represented on the number line.

## Linear equations in two variables

- An equation which can be expressed in the form  $ax + by + c = 0$ , where  $a$ ,  $b$  and  $c$  are real numbers such that  $a$  and  $b$  are not zero, is called a linear equation in two variables  $x$  and  $y$ .
- In the linear equation  $ax + by + c = 0$ ,  $a$  is coefficient of  $x$ ,  $b$  is coefficient of  $y$  and  $c$  is constant. Here  $a$ ,  $b$ ,  $c$  are called the coefficients of the linear equation.

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## Solution of linear equations in two variables

Solution of linear equation in two variables i.e.  $ax + by + c = 0$  has a pair of values, one for the unknown variable  $x$  and one for the second unknown variable  $y$  which satisfy the given equation.

**For eg.**

$2x + 3y = 6$ . When we substitute  $x = 1, y = 4/3$  in the given equation, we find that  $2 \times 1 + 4/3 \times 3 = 2 + 4 = 6$ . Therefore  $(1, 4/3)$  is a solution of the linear equation.

## Graph of a linear equation in two variables

In order to draw the graph of a linear equation  $ax + by + c = 0$ ,  $a \neq 0, b \neq 0$ , we may follow the following algorithm.

**Step-I :** Obtain the linear equation and let the axis equation be  $ax + by + c = 0$ .

**Step-II :** Express  $y$  in terms of  $x$  to obtain  $y = -\left(\frac{ax + c}{b}\right)$

**Step-III :** Put any two or three values of  $x$  and calculate the corresponding values of  $y$  from the expression in step-II to obtain two solution say  $(a_1, b_1)$  and  $(a_2, b_2)$ , if possible take values of  $x$  as integers in such a manner that the corresponding values of  $y$  are also integers.

**Step-IV :** Plot points  $(a_1, b_1)$  and  $(a_2, b_2)$ .

**Step-V :** Join the points marked in step IV to obtain a line.

The line obtained is the graph of the equation  $ax + by + c = 0$ .

## Equations of Lines Parallel to the x-axis and y-axis

$x = a$  is an equation of line parallel to y-axis

and  $y = b$  is an equation of line parallel to x – axis.

**For eg.**

Represent  $x - 5 = 0$  on the Cartesian plane.

**Sol.** We solve  $x - 5 = 0$ ;  $x = 5$ ; Here, the graph AB is line parallel to the y-axis and at a distance of 5 units to the right of it.

